S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 09

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Name: Suhani Birkhede

USN: CD24064

Semester / Year:4th/2nd

Academic Session:

Date of Performance:

Date of Submission:

CODE :-

```
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > farthest) {
                        farthest = next;
                        pos = j;
                    }
                }

                frames[pos] = pages[i];
            }
        }
    }

    printf("Total Page Hits = %d\n", hit);
    printf("Total Page Faults = %d\n", fault);
    printf("Hit Percentage = %.2f%%\n", (hit*100.0)/n);
    printf("Fault Percentage = %.2f%%\n", (fault*100.0)/n);

    return 0;
}
```

```
#include <stdio.h>

int main() {
    int frames[3], pages[] = {1,2,3,4,2,5,1,2,3,4,5};
    int n = 12, i, j, k, pos, farthest;
    int hit = 0, fault = 0;

    for(i = 0; i < 3; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check page hit
        for(j = 0; j < 3; j++) {
            if(frames[j] == pages[i]) {
                found = 1;
                hit++;
                break;
            }
        }

        if(!found) {
            fault++;

            // Check empty frame
            int empty = -1;
            for(j = 0; j < 3; j++) {
                if(frames[j] == -1) {
                    empty = j;
                    break;
                }
            }

            if(empty != -1) {
                frames[empty] = pages[i];
            }
            else {
                farthest = -1;
                pos = -1;

                for(j = 0; j < 3; j++) {
                    int next = -1;

                    for(k = i + 1; k < n; k++) {
                        if(frames[j] == pages[k]) {
```

OUTPUT:-

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ nano opt.c

Student@DESKTOP-JJ8CFNB MINGW64 ~
$

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ gcc opt.c -o opt

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ ./opt
Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ |
```